ASSIGNMENT - 1

Fundamentals of Data Science

Preprocessing Techniques

(1)

Student ID	Name	Age	Gender	Marks	Attendance	City
101	Ravi	20	M	85	90	Chennai
102	Priya	19	F	92	95	Chennai
103	Arjun	NULL	M	78	88	Madurai
104	Meena	20	F	450	91	Chennai
105	Ravi	20	M	85	90	Chennai
106	Kiran	21	Male	NULL	NULL	Coimbatore

(2)

EmpID	Name	Age	Gender	Department	Salary	Experience	Performance	City
E101	Arun	25	Male	IT	35,000	2	78	Chennai
E102	Divya	28	Female	HR	43,000	5	85	Madurai
E103	Karthik	27	M	IT	34,000	1	73	Chennai
E104	Priya	NULL	Female	Finance	98,000	6	91	Coimbatore
E105	Naveen	31	Male	IT	NULL	7	88	Coimbatore
E106	Hema	27	F	HR	80,000	4	82	Madurai
E107	Rahul	29	Male	Finance	1,50,000	5	87	Chennai
E108	Anitha	26	Female	IT	38,000	NULL	80	Coimbatore
E109	Sanjay	30	Male	HR	44,000	6	150	Madurai
E110	Kaviyu	25	Female	IT	36,000	2	76	Chennai
E111	Arun	25	Male	IT	35,000	2	78	Chennai
E112	Lakshmi	28	Female	Finance	43,000	4	84	Madurai

OUTPUT :

① Data Profiling

<class 'pandas.DataFrame'>

Range Index : 6 entries, 0 to 5

Data columns (total of columns):

#	Column	Non-Null Count	Dtype
0	Student-ID	6 non-null	int 64
1	Name	6 non-null	str
2	Age	5 non-null	float 64
3	Gender	6 non-null	str
4	Marks	5 non-null	float 64
5	Attendance	5 non-null	float 64
6	City	6 non null	str

dtypes : float 64 (3), int 64 (1), str (3)

memory usage : 550.0 bytes

Student ID 0
Name 0
Age 0
Gender 0
Marks 1
Attendance 1
City 0
dtype : 64

Data Profiling (Step 1):

1. Structure discovery: Inspecting schema and Data types Target rows and columns. All 12 rows across all columns.

Python Code:

df.info()

Function name: DataFrame.info()

Missing value: Age, number Attendance have 1 null value each

Inconsistent Gender: M, F, male.

Outlier Marks: 450 for students 109

Step 2: Data Cleansing

df['Gender'] = df['Gender'].replace({'M': 'male', 'F': 'Female'})

Constraint validation and Outlier Handling (Step 3):

df.loc[df['marks'] > 100, 'marks'] = np.nan

Step 4: Missing Data imputation

df['Age'] = df['Age'].fillna(df['Age'].median())

$$\text{mean} = \frac{85 + 92 + 78 + 85}{4} = 85$$

df['marks'] = df['marks'].fillna(df["marks"].mean())]

Attendance

Valid values: 90, 95, 88, 91, 90

median = 90

Student 106 Attendance = 90

df['Attendance'] = df['Attendance'].fillna(df[Attendance].median())

Step 5: Data Transformation

df["Academic_Score"] = df['Marks']*0.7 + df['Attendance']*0.3])

Age group

df['Age_group'] = pd.cut(df['Age'], bins=[0, 20, 100], labels=['Young', 'Adult',])

Data cleansing

	Student ID	Name	Age	Gender	Marks	Attendance	City
0	101	Ravi	20.0	Male	85.0	90.0	Chennai
1	102	Priya	19.0	Female	92.0	95.0	Chennai
2	103	Arjun	NULL	Male	78.00	88.0	Madurai
3	104	Meena	20.0	Female	450.0	91.0	Chennai
4	105	Ravi	20.0	Male	85.0	90.0	Chennai
5	106	Kiran	21.0	Male	NULL	NULL	Coimbator

After data handling constraints :

	Standent ID	Name	Age	Gender	Marks	Attendance	City
0	101	Ravi	20.0	Male	85.0	90.0	Chennai
1	102	Priya	19.0	Female	92.0	95.0	Chennai
2	103	Arjun	NULL	Male	78.0	88.0	Madurai
3	104	Meena	20.0	Female	NULL	91.0	Chennai
4	105	Ravi	20.0	Male	85.0	90.0	Chennai
5	106	Kiran	21.0	Male	NULL	NULL	Coimbatore

After missing data Imputation

	Student ID	Name	Age	Gender	Marks	Attendance	City
0	101	Ravi	20.0	Male	85.0	90.0	Chennai
1	102	Priya	19.0	Female	72.0	95.0	Chennai
2	103	Arjun	20.0	Male	78.0	88.0	Madurai
3	104	Meena	20.0	Female	85.0	91.0	Chennai
4	105	Ravi	20.0	Male	85.0	90.0	Chennai
5	106	Kiran	21.0	Male	85.0	90.8	Coimbatore

Data Transformation (Academic Score, Region, Age group)

	Student ID	Name	Age	City	Region	Agademenic Score
0	101	Ravi	20.0	Chennai	North TN	86.5
1	102	Priya	19.0	Chennai	North TN	92.90
2	103	Arjun	20.0	Madurai	South TN	81.00
3	104	Meena	26.0	Chennai	North TN	86.00
4	105	Ravi	20.0	Chennai	North TN	86.50
5	106	Kiran	21.0	Coimbatore	West TN	86.71

Data Reduction

	StudentID	Name	Marks
0	101	Ravi	85.00
1	102	Priya	92.00
3	104	Meena	85.00
4	105	Ravi	86.0
5	106	Kiran	85.0

Data Enrichment

	StudentID	Name	Age	Gender	Marks	Attendance
0	101	Ravi	20.0	Male	85.0	90.0
1	102	Priya	19.0	Female	92.0	95.0
2	104	Meena	20.0	Female	85.0	91.0
3	105	Ravi	20.0	Male	85.0	90.8
4	106	Kiran	21.0	Male	85.0	90.8

	City	Academic score	Region	Age group	City index
0	Chennai	86.50	North-TN	19.20	1.2
1	Chennai	92.90	North TN	19.00	1.2
2	Chennai	86.80	North TN	19.00	1.2
3	Chennai	86.50	North TN	19.00	1.2
4	Coimbatore	80.74	West TN	21.00	1.0

Employee table

Step 1 : Data Profiling

Problem Identified :

E104 → Age = NULL

E105 → Salary = NULL

E108 → Experience = NULL

Gender has M, male, Male, Male, F, Female, FEMALE

Department has finance and Finance

E107 → salary = 45000

E109 → Performance = 150

E101 and E111 → duplicate records

Step 2 : Data Cleansing

```
df['Gender'] = df['Gender'].replace({'M':'Male', 'male':'Male', 'F':'Female',
                                     'FEMALE':'Female'})
df['Department'] = df['Department'].replace({'finance':'Finance'})
```

Salary

```
df['salary'] = (df['salary'].str.replace(',', '').astype(float)
```

Step 3 : Constraint validation and outlier handling performance

Should be between 0 and 100.

E109 had 150 → invalid

```
df.loc[df['Performance'] > 100, 'Performance'] = np.nan
```

Salary outlier

Q_1 = df['salary'].quantile(0.24)

Q_2 = df['salary'].quantile(0.76)

IQR = $Q_3 - Q_1$

Upper_limit = $Q_3 + 1.5 * IQR$

```
df['salary'] = df['salary'].clip(upper=upper_limit)
```

OUTPUT

Data Profiling

<class 'pandas.Dataframe'>

Range Index : 12 entries (0 to 11

Data Columns (total 9 columns) :

#.	Column	non-null count	Dtype
0	Emp_id	12 non-null	Str
1	Name	12 non-null	Str
2	Age	11 non-null	float64
3	Gender	12 non-null	Str
4	Department	12 non null	Str
5	Salary	12 non null	Str
6	Experience	11 non null	float64
7	Performance	12 non null	int64
8	City	12 non null	Str

dtypes : float64(2), int64(1), Str(6)

memory usage : 1.3 kB

Emp ID	0
Name	0
Age	1
Gender	0
Department	0
Salary	1
Experience	1
Performance	0
City	0

dtype : int64

Data Cleansing

	Emp_ID	Name	Age	Gender	Department	Salary	Experience	Performance	City
0	E101	Arun	25.0	Male	IT	35,000	2	78	Chennai
1	E102	Divya	28.0	Female	Hr	44,000	5	85	Madurai
2	E103	Karthik	24.0	Male	It	34,000	1	72	Chennai
3	E104	Priya	NULL	Female	Finance	98,000	6	91	Coimbatore
4	E105	Naveen	31.0	Male	IT	NULL	7	88	Coimbatore
5	E106	Meena	27.0	Female	Hr	40,000	4	82	Madurai
6	E107	Rahul	29.0	Male	Finance	4,50,000	5	87	Chennai
7	E108	Anitha	26.0	Female	IT	38,000	NULL	80	Coimbatore
8	E109	Sanjay	30.0	Male	Hr	44,000	6	150	Madurai
9	E110	Kavya	25.0	Female	It	34,000	2	76	Chennai
10	E111	Arun	25.0	Male	It	35,000	2	78	Chennai
11	E112	Lakshmi	28.0	Female	Finance	43,000	4	84	Madurai

Constraint validation :

	EmpID	Name	Age	Gender	Dept	Salary	Experience	Performance	City
0	E101	Arun	25	Male	IT	35,000	2	78	Chennai
1	E102	Divya	28	Female	HR	40,000	5	85	Madurai
2	E103	Karthik	27	Male	IT	34,000	1	72	Chennai
3	E104	Priya	NULL	Female	Finance	98,000	6	91	Coimbatore
4	E105	Naveen	31	Male	IT	NULL	7	88	Coimbatore
5	E106	Meena	27	Female	HR	90,000	4	8.2	Madurai
6	E107	Rahul	29	Male	Finance	4,50,000	5	87	Chennai
7	E108	Anitha	26	Female	IT	38,000	NULL	80	Coimbatore
8	E109	Sanjay	30	Male	HR	44,000	6	NULL	Madurai
9	E110	Kaviya	25	Female	IT	34,000	2	78	Chennai
10	E111	Arun	25	Male	IT	35,000	2	78	Chennai
11	E112	Lakshmi	28	Female	Finance	43,000	4	84	Madurai

Missing Data Imputation

	EmpID	Name	Age	Gender	Department	Salary	Experience	Performance	City
0	E101	Arun	25.0	Male	IT	35,000	2	78	Chennai
1	E102	Divya	28.0	Female	HR	42,000	5	85	Madurai
2	E103	Karthick	27.0	Male	IT	34,000	1	72	Chennai
3	E104	Priya	29.0	Female	Finance	98,000	6	91	Coimbatore
4	E105	Naveen	31.0	Male	IT	40,000	7	88	Coimbatore
5	E106	Meena	27.0	Female	HR	40,000	4	82	Madurai
6	E107	Rahul	29.0	Male	Finance	4,50,000	5	87	Chennai
7	E108	Anitha	26.0	Female	IT	38,000	4	80	Coimbatore
8	E109	Sanjay	30.0	Male	HR	44,000	6	150	Madurai
9	E110	Kaviya	25.0	Female	IT	36,000	2	76	Chennai
10	E111	Arun	25.0	Male	IT	35,000	2	78	Chennai
11	E112	Lakshmi	28.0	Female	Finance	43,000	4	84	Madurai

Data Transformation:

	Emp ID	Name	Age	City	Age Group	Region
0	E101	Arun	25.0	Chennai	young	North TN
1	E102	Divya	28.0	Madurai	young	South TN
2	E103	Karthick	27.0	Chennai	young	North TN
3	E104	Priya	27.0	Coimbatore	young	West TN
4	E105	Naveen	31.0	Chennai	Middle Age	North TN
5	E106	Meena	27.0	Madurai	young	South TN
6	E107	Rahul	29.0	Chennai	Middle Age	North TN
7	E108	Anitha	26.0	Coimbatore	young	West TN
8	E109	Sanjay	30.0	Madurai	Middle Age	South TN
9	E110	Kaviya	25.0	Chennai	young	North TN
10	E111	Arun	25.0	Chennai	young	North TN
11	E112	Lakshmi	28.0	Madurai	young	South TN

Data Normalisation

0	0.315789
1	0.684211
2	0.000000
3	1.000000
4	0.842105
5	0.526316
6	0.789474
7	0.421053
8	0.526316
9	0.210526
10	0.315789
11	0.631579

Data Enrichment

	Age	Salary	City	City_Index
0	25	35,000	Chennai	1.2
1	28	42000	Madurai	1.0
2	24	32000	Chennai	1.2
3	27	48000	Coimbatore	0.7

ASSIGNMENT - 2

EDA

From the given table

Step 1 : Import libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2 : Load dataset

```
df = pd.read_csv("dataset.csv")
```

Step 3 : Display first 5 rows

```
print(df.head())
```

Step 4 : Check dataset information

```
print(df.info())
```

Step 5 : Check shape

```
print(df.shape)
```

Step 6 : Display column names

```
print(df.columns)
```

Step 7 : Check missing values

```
print(df.isnull.sum())
```

Step 13 - Correlation heatmap:

```
sns.heatmap(df.select_dtypes("number").
                corr().annot=True)
plt.title("correlation Heatmap")
plt.show()
```

Output:

First 5 rows:

	State	bidprice	know	wtp	wtp1	wtp2	confident	gender	age
0									
1									
2									
3									
4									

Dataset shape

(rows, columns)

Missing values:

State	0
bidprice	0
know	0
wtp	0
wtp1	0

confident 0

gender 0

age 0

race 0

marital 0

edu 0

edu 2 0

work 0

whsize 0

kids 0

Income 0

eduyear 0

Duplicate Rows

0

Satisfied Summary :

	age	whsize	kids
income			
count			
mean			
std			
min			
25%			
50%			
75%			

ASSIGNMENT - 3

11 Marks

(1) a) * A sudden 40% increase in bounce rate after a website redesign is a warning sign that user experience or website performance may have been affected.

① Investigate bounce rate first

* Compare the bounce rate before and after the redesign using the same time period and traffic sources.

* Check whether the increase is happening across the entire website or only on specific pages, devices, browsers, or regions.

* Identify pages with the highest bounce-rate increase, especially important landing and product pages.

② Check website and technical issues

* Investigate page-load speed, broken links, mobile responsiveness, JavaScript errors, and checkout problems.

* Compare the new design with the old design to identify changes that may be causing users to leave.

* Verify that analytics tracking is working correctly because an implementation error can also produce an unusual bounce-rate change.

③ Analyse customer segments

* Divide customers into new visitors, repeat buyers, high-value customers, and inactive customers.

* Check whether new customers are leaving more frequently while repeat customers are spending more.

* Analyse conversion rate, average order value, purchase frequency, and customer retention for each segment.

④ Investigate why revenue stayed flat

Revenue can be represented as:

Revenue = Number of customers x Conversion Rate x Average Order Value

* A fall in customers or conversions may be compensated by higher spending from repeat buyers.

* Therefore, I would investigate whether revenue is becoming dependent on a small customer segment, which may be risky in the long term.

⑤ Analyse user behaviour:

* Use funnel analysis to find where users leave:

Landing Page → Product Page → Cart → Checkout → Purchase

* Compare user behaviour before and after the redesign.

* Conduct A/B testing or user feedback analysis to determine whether the redesign is actually causing poor engagement.

⑥ ML Engineer:

I would involve the ML Engineer for customer segmentation and reports showing bounce rate, conversion, and revenue trends.

⑦ BI Analyst:

I would involve the BI Analyst to create dashboards and reports showing bounce rate, conversion, and revenue trends.

⑧ CDO:

I would involve the CDO for business strategy, data governance, and decision-making.

Classification and Characteristics of Different Types of Data.

1. Structured Data

* Data stored in a fixed format such as rows and columns.
* Easy to store, search and analyse.
* Example : Student database, sales records.

2. Semi-Structured Data

* Does not follow a strict table format but has tags or keys.
* More flexible than structured data.
* Example : JSON, XML, emails.

3. Unstructured Data

* Has no predefined structure.
* Requires special techniques for analysis.
* Example : Images, videos, audio, social media posts.

4. Qualitative Data

* Describes qualities or categories rather than numbers.
* Used to understand opinions, behaviour and characteristics.
* Example : Gender, colour, customer feedback.

5. Quantitative Data

* Represents numerical values that can be measured.
* Can be used for mathematical and statistical analysis.
* Example : Age, salary, height, marks.

6. Discrete Data :

* Consists of countable values, usually whole number.
* Example : Number of students, number of products so

7. Continuous Data :

* Can take any value within a range.
* Example : Height, weight, temperature.

I would choose Hospital and Bank because their goals and data are very different.

1. Hospital - Reduce Readmissions

Goal : Predict which patients are likely to be readmitted.

Type of Data :

* Patient age, medical history, diagnosis, medicines, and treatment records.
* Mostly structured data, along with unstructured doctor notes.
* Data is sensitive and requires strong privacy protection.

Analysis Approach :

* Clean and preprocess patient data.
* Identify factors related to readmission.
* Use classification and predictive models to identify high-risk patients.
* Evaluate the model using accuracy, precision, recall, etc.
* Doctors can use the results to provide additional care to high-risk patients.

1. Security Issues.

a) During Data Collection

* Collection unnecessary personal or sensitive information can increase security risks.

* Fake or incorrect data may enter the system through insecure sources.

b) During Data Storage

* Data breaches can expose confidential information.

* Weak passwords and poor access controls can allow unauthorized users to access data.

* Malware and ransomware can damage or lock stored data.

c) During Data Processing:

* Unauthorized employees or applications may misuse data.

* Malware can modify or corrupt data during processing.

d) During Data sharing

* Data can be intercepted while being transferred.

* Third-party services may accidentally expose confidential information.

* sharing can lead to privacy violations

2. Bank - Detect Fraud

Goal: Identify suspicious transactions.

Type of Data:

* Transaction amount, location, time, account details, and transaction history.
* Mostly structured and numerical data.
* Large amounts of transaction data are generated continuously.

Analysis Approach:

* Analyse normal and suspicious transaction patterns.
* Use classification, anomaly detection, and machine learning
* Identify unusual transactions and generate fraud alerts.
* Evaluate the model using precision, recall, and false-positive rate.

Difference:

Hospital	Bank
Patient and medical data	Transaction data.
Predicts readmission risk	Detects fraudulent activity
Classification / prediction	Classification / anomaly detection.
Focus on patient safety.	Focus on financial security.
Strong medical privacy required	Strong financial security required.

easures to Protect Data:

1) Encryption - Protect data from unauthorized access.
2) Strong Passwords - Use strong passwords and MFA.
3) Access Control - Allow access only to authorised users.
4) Regular Backups - Prevent data loss.
5) Firewall & Antivirus - Protect against malware and attacks.
6) Software Updates - Fix security vulnerabilities.
7) Employee Training - Tech employees about phishing and securing Threats.
8) Security Audits - Regularly check and monitor security.

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① Identify the Data Required:

* Teaching method used: Lecture, Group Project, Flipped Classroom etc.
* Student marks and attendance.
* Student engagement and satisfaction.

② Select Participants:

* Include students and teachers from different department and years.
* Use a representative sample to avoid bias.

③ Data Collection Methods:

* Questionnaries.
* Interviews.
* Academic records
* Classrom observation

④ Ensure Fair Comparisson

* Collect data over the same academic period.
* Compare similar subjects and student groups where possible.

* Use the same questionnaire and measurement criteria across departments.

⑤ Data Storage and Privacy:

* Store the collected data securely
* Remove personal identifiers and maintain student privacy.

⑥ Analyse the Data

* Compare average marks, attendance, engagement, and satisfaction for each teaching method.
* Identify which methods performs best in each department.

Steps:

① Data Cleaning:

* Identify missing values such as Age for P101 and missing Blood Pressure for P104.
* Fill missing values using suitable methods such as mean/median for numerical data for mode for categorical data.

② Remove Duplicates

* P103 appears twice with the same information.
* Keep only one P103 record.

③ Correct Inconsistent Formats

* Convert Gender values "male" → "Male"
* Convert Hospital names such as "gh" and "GH" into one standard format.
* Standardize "NORMAL" and "Govt Hospital" according to the correct hospital name.

* Use the same questionnaire and measurement criteria across departments.

5) Data Storage and Privacy:

* Store the collected data securely
* Remove personal identifiers and maintain student privacy.

6) Analyse the Data

* Compare average marks, attendance, engagement, and satisfaction for each teaching method.
* Identify which methods perform best in each department.

Steps:

1) Data Cleaning:

* Identify missing values such as Age for P101 and missing Blood Pressure for P104.
* Fill missing values using suitable methods such as mean/median for numerical data for mode for categorical data.

2) Remove Duplicates

* P103 appears twice with the same information.
* Keep only one P103 record.

3) Correct Inconsistent Formats:

* Convert Gender values "male" → "Male"
* Convert Hospital names such as "gh" and "GH" into one standard format.
* Standardize "NORMAL" and "Govt Hospital" according to the correct hospital name.

⑤ Correct Invalid Values :

* "1280/82" is likely an incorrectly entered blood-pressure value and should be verified and corrected.

* Standardize blood pressure as systolic / diastolic.

⑤ Handle Irrelevant Attributes

* Remove attributes that are not required for the analysis.

* Keep useful attributes such as Patient ID, Age, Gender, Blood Pressure, Hospital, and Diagnosis.

⑥ Data Transformation :

* Convert Age into a consistent numerical format.

* Encode categorical values such as Gender and Diagnosis if required by the machine-learning algorithm.

⑦ Validation :

* Check that all patient IDs are unique.

* Check that ages and blood-pressure values are within reasonable ranges.

* Verify corrected values with the original hospital records.

---

14) a) Data Cleaning Approach :

* Missing Values :

* Issue : Age for C102 and Purchase_Amount for C104 contain missing values (-).

* Approach : Replace dashes with NaN values. Impute the missing values using the column mean or median.

* Duplicate Record:

* Issue: The row of customer C103 is repeated identically.

* Approach: Identify and remove duplicate rows to keep only the first occurrence.

* Inconsistent Formatting:

* Issue: Mixed casing exists in Gender ("Female" vs "female") and City ("Chennai" vs "chennai").

* Approach: Standardize string formatting by converting text columns to title case or lowercase.

* Likely Data-Entry Error

* Issue: Customer C105 has a Purchase Amount of 410000, which is an extreme outlier compared to others (~3,000 to 5,000)

* Approach: Filter out the outlier using threshold filtering (eg IQR method) or manually cap/correct the value.

Python code:

```
import pandas as pd
import numpy as np
data = {
    'Customer_ID' : ['C101', 'C102', 'C103', C103', 'C104', 'C105'],
    'Age' : [25, np.nan, 31, 31, 28, 35],
    'Gender' : ['Female', 'Male', 'female', 'female', 'Male', 'Male'],
    'City' : ['Chennai', 'Chennai', 'Chennai', 'Chennai', 'Madurai', 'chennai'],
    'Purchase_Amount' : [4500, 3200, 5600, 5600, np.nan, 410000]
}
df = pd.DataFrame(data)
df = df.drop_duplicates(keep='first')
df['Gender'] = df['Gender'].str.title()
df['City'] = df['City'].str.title()
print(df)
```

① Numerical Attributes:

* Year, Budget, Rating and Duration.

② Technique:

* Use Equal-Width Binning because it divides the range into equal intervals and is simple for this small dataset.

③ Example - Budget

Budget (₹ Cr)	Category
15 – 36.7	Low
36.7 – 58.3	Medium
58.3 – 80	High

- Thus, 25 - Low, 40 → Medium, 60 - High

④ Python Code:

```
data['Budget_Category'] = pd.cut(
    data['Budget'], bins = 3,
    labels = ['Low', 'Medium', 'High']
)
```

⑤ Usefulness:

* Simplifies numerical data
* Makes comparison easier.
* Helps in classification and further analysis.

15) b) * EDA is used to understand patterns, trends, and important observation in a dataset

① Data Understanding:

* The data contains 12 months and four product categories:
  * Electronics
  * Clothing
  * Groceries
  * Furniture

② Data Cleaning:

Check for missing values, duplicates, and incorrect entries. The given data has no obvious missing values.

③ Descriptive Analysis:

Calculate total, average, minimum and maximum sales for each category.

④ Trend Analysis: Sales generally increase from January to December.

⑤ Visualization: Use line charts and bar charts to compare sales.

⑥ key findings:

* Electronics has the highest sales.
* Furniture has the lowest sales.
* December has the highest sales for all categories.
* Sales increase significantly during November and December.

ASSIGNMENT - 4

UNIT - 4

Two - independent sample t-test

Given :

Treatment mean = 26.4

Control mean = 18.6

Standard mean = 2.4

$n = 120$

$df = 118$

$\alpha = 0.05$

Critical $t = \pm 1.980$

Hypothesis test

$H_0 : \mu_1 - \mu_2 = 0$, $H_1 : \mu_1 - \mu_2 \neq 0$

$$t = \frac{26.4 - 18.6}{2.4}$$

$t = 3.25$

$t = 3.25$

table value $= \pm 1.980$

Since $3.25 > 1.980$

Reject H_0

Conclusion : There is a statistically significant difference in driving error between two groups.

P - value :

$P \approx 0.0015$

Since $p < 0.05$, the result is statistically significant.

iii) 95% Confidence interval

$$26.4 - 18.6 = 7.8$$

$$C_1 = 7.8 \pm (1.980)(2.4)$$

$$C_1 = 7.8 \pm 4.752$$

$$(3.048, 12.552)$$

The population mean difference is estimated to be between 3.05 and 12.55 errors.

iv) Cohen's d:

$$S_p = 13.15$$

$$d = \frac{26.4 - 18.6}{13.15} = 0.593$$

This indicates moderate effect size.

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② ANNOVA:

Given:

Tenderiser A: 36, 28, 42, 58

Tenderiser B: 17, 59, 33

Tenderiser C: 36, 74, 29, 55, 48

$\alpha = 0.05$

Group A:

36, 28, 42, 58

Total = $36 + 28 + 42 + 58 = 164$

$n = 4$

$$\text{mean} = \frac{164}{4} = 41.$$

Group B:

17, 59, 33

Total = $17 + 59 + 33 = 109$

$n = 3$.

mean $= \frac{109}{3} = 36.33$

Group C :

36, 74, 29, 55, 48

Total $= 36 + 74 + 29 + 55 + 48 = 242$

$n = 5$

mean $= \frac{242}{5} = 48.4$

Grand Total $= 164 + 109 + 242$

$T = 515$

total observation $= 4 + 3 + 5$

$N = 12$

Grand mean

$\bar{X} = \frac{515}{12} = 42.917$

$SSB = \sum \frac{T_i^2}{n_i} - \frac{T^2}{N}$

$= \frac{164^2}{4} + \frac{109^2}{3} + \frac{242^2}{5} - \frac{515^2}{12}$

$SSB = 41.67$

$\sum x^2 = 36^2 + 28^2 + 43^2 + 58^2 + 17^2 + 99^2 + 33^2 + 36^2 + 74^2 + 29^2 + 55^2 + 48^2$

$\sum x^2 = 23855$

$SST = \sum x^2 - T^2/N$

$= 23985 - \frac{515^2}{12}$

$= 23985 - 22102.08$

$= 1873.92$

$SSW = SST - SSB$

$= 1873.92 - 41.67$

$SSW = 18.32.25$

$df = N-1 \quad = 12-1$

$= 11$

$MSB = \frac{SSB}{df_1} = \frac{41.67}{2} = 20.83$

$MSW = \frac{SSW}{df_2} = \frac{2018.25}{9} = 203.58$

$F = \frac{MSB}{MS_W} = \frac{20.835}{203.583}$

$\boxed{F = 0.102}$

Given $F_{0.05}(2,9) = 4.26$

$F = 0.102$

Since $0.102 < 4.26$

accept H_0

There is no significant difference in the mean weight loss of meat between the three tenderisers at the 5% significance level.